

Homework 8

Solutions must be **typed** and **submitted electronically**.

- Let F be a secure PRP with blocklength λ . Consider the following encryption mode:

$$\begin{array}{l} \text{Enc}(k, m_1 \| \dots \| m_\ell): \\ \quad c_0 \leftarrow \{\mathbf{0}, \mathbf{1}\}^\lambda \\ \quad \text{for } i = 1 \text{ to } \ell: \\ \quad \quad c_i := F(k, m_i) \oplus c_{i-1} \\ \quad \text{return } c_0 \| \dots \| c_\ell \end{array}$$

- Write the corresponding decryption algorithm.
 - Show that the mode does not achieve CPA\$ security.
- Implementers are sometimes cautious about IVs in block cipher modes and may attempt to “protect” them. One idea for protecting an IV is to prevent it from directly appearing in the ciphertext. The modified CBC encryption below sends the IV through the block cipher before including it in the ciphertext:

$$\begin{array}{l} \text{Enc}(k, m_1 \| \dots \| m_\ell): \\ \quad c_0 \leftarrow \{\mathbf{0}, \mathbf{1}\}^{blen} \\ \quad c'_0 := F(k, c_0) \\ \quad \text{for } i = 1 \text{ to } \ell: \\ \quad \quad c_i := F(k, m_i \oplus c_{i-1}) \\ \quad \text{return } c'_0 \| c_1 \| \dots \| c_\ell \end{array}$$

This ciphertext can be decrypted by first computing $c_0 := F^{-1}(k, c'_0)$ and then doing usual CBC decryption on $c_0 \| \dots \| c_\ell$.

Show that this new scheme is **not** CPA-secure.